Adversarial Attacks and Defenses on Large Language Models: A Systematic Review

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***Abstract*—The rapid advancement of large language models (LLMs) presents both transformative opportunities and significant security challenges, primarily due to their susceptibility to adversarial attacks—crafted inputs designed to deceive them—which pose considerable risks to their reliability and security. This survey comprehensively reviews these textual adversarial attacks, categorizing them into sentence-level, word-level, and character-level manipulations, while also exploring multi-level approaches that combine these techniques. To counter such threats, the survey examines a range of defense strategies, including detection methods, adversarial training, semantic analysis, general system enhancements, and approaches to certified robustness for provable resilience. Despite notable progress in understanding and mitigating these vulnerabilities, significant challenges remain regarding real-world applicability, scalability, generalization across different models and attack types, and the establishment of standardized benchmarking. By synthesizing recent research in this dynamic field, this survey aims to guide the development of more secure and robust language models capable of withstanding the continuously evolving landscape of adversarial threats.**

***Index Terms*—Textual Adversarial Attack, Sentence-level Attack, Word-level Attack, Char-level Attack, Multi-level Attack, Texture Model Defence, Certified Robustness.**

I. Introduction

In recent years, the rapid advancement of large language models (LLMs) have revolutionized the field of natural language processing (NLP), enabling unprecedented capabilities in tasks such as financial [1], [2], recommdation [3], text generation [4]–[9], sentiment analysis [10]–[15], medical [16]–[19] and biological areas [20]–[24]. However, as these models become increasingly sophisticated, their susceptibility to adversarial attacks—carefully crafted inputs designed to deceive or mislead the model—also grows [25], [26]. Such attacks pose significant risks to the reliability and security of NLP systems, potentially leading to incorrect predictions, biased outputs, or even malicious exploitation [27]–[31].

This survey provides a comprehensive review of current research on textual adversarial attacks and defense mechanisms for large language models. Adversarial attacks are categorized into three primary levels based on the granularity of linguistic manipulation: sentence-level, word-level, and character-level attacks [32]–[34], with each attack category explored in depth regarding its methodologies, techniques, and key contributions. Additionally, the survey examines multi-level attacks that combine these approaches to create more sophisticated adversarial examples [35]–[37]. On the defense side, a structured overview of strategies to enhance NLP model robustness is presented. These defense mechanisms are grouped into four main categories: detection and identification, adversarial training and mitigation, semantic and text analysis, and general enhancement techniques [38]–[40]. Each defense category is discussed in detail, highlighting the strengths, limitations, and real-world applicability of various approaches [41], [42]. Furthermore, the survey delves into certified robustness in NLP, which offers formal guarantees of a model's resilience to adversarial perturbations [43], [44], reviewing recent advancements in this area, including novel algorithms and frameworks for provable robustness against diverse attack types [1], [45], [46].

By synthesizing the latest advancements and highlighting critical challenges and opportunities, this survey serves as a comprehensive resource for researchers and practitioners dedicated to understanding and mitigating the risks posed by adversarial attacks in NLP. Its overarching objective is to catalyze the development of more secure, reliable, and resilient language models capable of navigating the ever-evolving landscape of adversarial threats.

II. Overview of Textual Adversarial Attack

A. Related Work on Sentence-level Attack (Done)

Sentence-level adversarial attacks manipulate entire sentences or phrases to mislead NLP models, thereby altering their outputs or guiding them toward incorrect predictions. These attacks often exploit strategies designed to confuse models, rendering them unable to process the input text correctly. For instance, Lin [47] revealed statistical biases in machine reading comprehension models, showing a tendency to favor irrelevant options over true understanding. Similarly focused on model vulnerabilities, Xu [48] proposed a grey-box framework for sentiment classification, using Gumbel-softmax and competitive objectives to generate high-quality adversarial examples. Generation techniques have also advanced: SynPG [49] presented syntax-controlled generation through semantic syntactic disentanglement, enhancing diversity without annotated data, while CAT-Gen [50] developed a model that generates diverse adversarial texts through precise control over task-irrelevant attributes, thereby improving output diversity and attack robustness. Ensuring semantic integrity, T3 implemented target-controllable attacks through tree-structured

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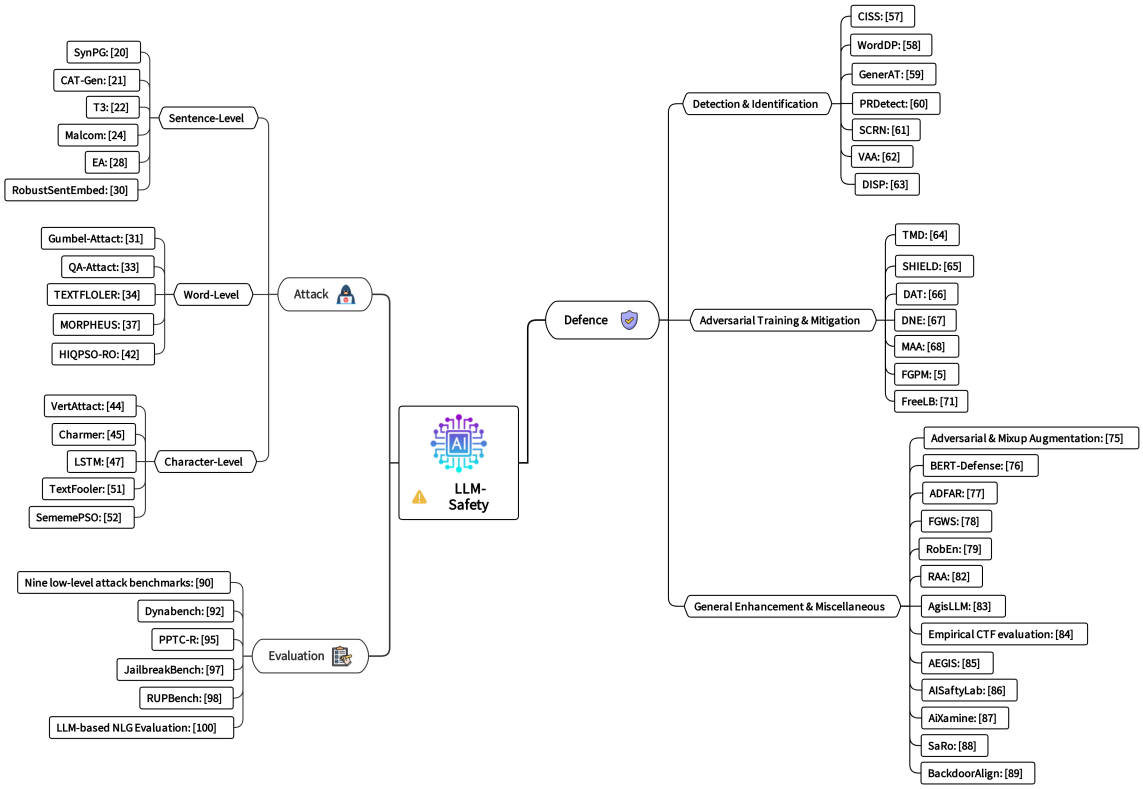

Fig. 1. The overall framework of this paper, centered on LLM Safety and branching into three key investigative areas: adversarial Attacks, mitigation Defences, and systematic Evaluation.

autoencoding, ensuring adversarial perturbations maintain text coherence and semantic consistency [51]. Other targeted approaches include work by Han [52], who attacked structured prediction models using a Seq2Seq model trained through reinforcement learning, and Malcom [53], which introduced an end-to-end framework that generates adversarial comments to attack fake news detection systems by optimizing perturbations while preserving semantic coherence. Human involvement has also been explored: Wallace [54] involved human authors in creating adversarial examples to test model weaknesses, and the PAWS dataset [55] was developed to measure model sensitivity to word order and structural variations through controlled perturbations. More recent black-box methods include an approach by Moraffah et al. [56] that draws on class probabilities to craft effective adversarial examples, while an edit-based adversarial text attack (EA) proposed by Li et al. [57] alters sentence structures, adds definitions, or inserts phrases to significantly boost attack success rates. Furthermore, Gu et al. [58] introduced a sentence-level inference attack targeting pre-trained language models by training decoders to reconstruct text from embeddings, demonstrating practical adversarial threats; in response, RobustSentEmbed [59] was developed to enhance sentence embedding robustness through adversarial self-supervised contrastive learning to defend against such attacks. These varied methods highlight the evolving sophistication of sentence-level attacks, emphasizing the critical need for robust defenses in NLP systems..

B. *Related Work on Word-level Attack*

Word-level adversarial attacks, which modify individual words to deceive NLP models while preserving text meaning and grammar, exploit model vulnerabilities through subtle, often human-imperceptible changes. For instance, Gumbel Attack [60] optimizes objectives via parameterized distributions and the Gumbel trick, aiming to reduce or eliminate the need for model evaluation during attacks (a challenge also addressed by methods like Seq2Sick for sequence models [61]). This approach also categorized attacks into non-overlapping and targeted keyword types using distinct loss functions for seq2seq models. Another approach, QA-Attack [62], combines the attention mechanism with a deletion ranking method to execute successful word-level adversarial attacks in question-answering models while maintaining semantic consistency. TEXTFLOLER [63] introduced a two-stage framework that ranks words by impact and then replaces them under semantic and grammatical constraints. Reinforcement learning has also been employed, for example, by Zou et al. [64], to iteratively degrade model performance in neural machine translation. Focusing on linguistic realism, grammar-based attacks simulating natural language errors such as typos and incorrect verb forms were developed by Yin et al. [65], while MORPHEUS [66] was proposed to perturb inflectional forms to uncover model biases and improve robustness. To test specific model capabilities, Shi et al. [67] modified shared vocabulary in sentence pairs for paraphrase identification models. Separately, Emelin et al. [68] introduced 'attractive vocabulary'

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to expose the reliance of neural machine translation (NMT) models on statistical features. Geometric approaches have also emerged, such as those by Meng et al. [69], utilizing saliency scores and proximity to decision boundaries to generate effective adversarial samples, while Maheshwary et al. [70] implemented a population-based optimization algorithm to maximize semantic similarity with minimal text modifications. More advanced optimization is seen in HIQPSO-RD [71], which further integrates the understanding of semantic primitives with an improved particle swarm optimization algorithm to generate adversarial samples achieving a high success rate at a low modification rate [72], [73]. These advancements collectively highlight the evolving landscape of word-level adversarial attacks, offering crucial insights into NLP model vulnerabilities and paving the way for the development of more robust systems.

C. Related Work on Char-level Attack (Done)

Character-level adversarial attacks manipulate individual characters to disrupt text meaning and deceive NLP models [32]. Early demonstrations of vulnerability involved introducing natural and synthetic noise, such as spelling mistakes and character shuffling, to character-based NMT models [74]. More targeted methods have since emerged. For example, VertAttack [75] was proposed as an adversarial method targeting text classifiers through vertical keyword rewriting, bypassing horizontal text processing mechanisms to induce significant accuracy degradation while preserving high human comprehension rates. Ebrahimi et al. [33] advanced letter-edit operations through vector representation and gradient estimation, employing strategies like one-shot, greedy algorithms, and beam search to optimize character replacements and significantly degrade translation quality. Similarly, Charmer [76], an efficient query-based character-level attack, generates highly similar adversarial examples with high success rates, effectively targeting models including BERT and Llama 2. Black-box approaches have also been developed; Gao et al. [77] proposed a method selecting words for modification based on model responses to word removal, achieving notable reductions in classification accuracy. In other investigations, Rocamora et al. [76] leveraged Chain-of-Thought (CoT) prompting techniques combined with character-level substitutions to investigate AraBERT model robustness, revealing that fine-tuned models exhibit greater resilience than original architectures. To enhance black-box attack efficiency, Gil et al. [78] employed LSTM networks to emulate white-box attacks without compromising success rates. Beyond typical character edits, Eger et al. [79] introduced a visual perturbator that swaps characters with visually similar counterparts, effectively evading sensitive word detection models through glyph-based substitution. Punctuation manipulations also prove effective: Du et al. [80] developed punctuation-level attacks combined with TPPE/TPPEP embedding search methods to systematically probe text model vulnerability to punctuation perturbations, demonstrating that single-punctuation modifications can deceive models while preserving human readability. Correspondingly, Formento et al. [81] explored punctuation insertion as an adversarial strategy, showing its superior performance over traditional character-level attacks like TextFooler [82] and SememePSO [83] in maintaining semantic similarity and readability. These diverse techniques highlight the significant vulnerabilities of NLP models to subtle character manipulations and underscore the critical need for robust defenses.

III. Overview of Textual Model Defense

In the rapidly evolving domains of network security and NLP, safeguarding text models against adversarial attacks has become paramount. Adversarial attacks, which involve meticulously engineered inputs, pose significant threats to model resilience and security [25]. These attacks are often categorized based on the attacker's knowledge of the target model and the granularity of linguistic operations involved [84]. To counter such threats, a comprehensive defense framework has emerged, encompassing diverse strategies to enhance model robustness. The first line of this defense involves detection and identification techniques; leveraging heuristics, statistical analysis, and anomaly detection, these methods flag adversarial inputs by identifying deviations from expected linguistic patterns [27], [28], serving as a foundational layer for targeted mitigation.

Building on detection, adversarial training and mitigation strategies constitute the second pillar, fortifying NLP models by exposing them to crafted adversarial examples during training. This exposure enhances generalization and resistance to manipulation [39]. Post-training techniques, such as gradient masking and input sanitization, further bolster this resilience by reducing susceptibility to known attack patterns [46]. A third crucial aspect of defense focuses on enhancing semantic and contextual understanding. Techniques like semantic knowledge graphs, contextual embeddings, and syntactic analysis help models differentiate legitimate inputs from adversarial ones; integrating such deep semantic understanding, often during the training phase, significantly improves overall model resilience [11], [85].

Finally, general enhancements such as ensemble methods, regularization techniques, and hybrid approaches provide supplementary protection, forming the fourth component of this framework [41], [42]. These strategies aim to address gaps left by more specific defenses, thereby offering additional layers of robustness. Together, these four pillars—detection, adversarial training, semantic analysis, and general enhancements—form a cohesive defense framework that advances the state of the art in text model security [26], [45], [86]. By integrating these multifaceted strategies, researchers can strive to develop resilient NLP systems capable of withstanding the increasingly sophisticated tactics employed in adversarial attacks.

A. Related Work on Detection and Identification (Done)

In the realm of detecting and identifying word-level adversarial text attacks, several innovative methods have emerged to bolster the security of natural language processing (NLP) systems. For instance, Shapley Additive Explanations [87] has been employed to efficiently pinpoint subtle, potentially malicious perturbations within text data, directly aiding in

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their recognition and subsequent mitigation. Establishing clear evaluation criteria is also crucial; an approach by Yoo and Qi [88] provides a robust benchmark for detecting adversarial examples in text classification, serving as a baseline to assess detector vulnerability and performance. Furthermore, methods like Causal Intervention for Sentence Substitution (CISS) [89] and WordDP [90] offer provable and certified robustness against word substitution attacks, which inherently enhances the reliability of detection mechanisms by providing clearer distinctions between benign and adversarial inputs. Some approaches integrate detection into the training process itself; the Generative Adversarial Training (GenerAT) framework [91], for example, aligns gradient-based token generation with perturbed token detection and symmetric KL regularization, thereby improving model robustness against detected manipulations. Advanced detectors also leverage linguistic structures; the Perturbation Robust detector (PRDetect) [92] utilizes syntactic tree features invariant under token-level disturbances to significantly enhance detection robustness with minimal latency. Similarly focused on robust detection, the Siamese Calibrated Reconstruction Network (SCRN) [93] denoises token embeddings and enforces prediction consistency via symmetric KL calibration; these actions strengthen resilience and improve the ability to detect adversarial attacks, especially across diverse AI-generated text benchmarks. Understanding vulnerabilities is key to detection, and the Vulnerability via Adversarial Attack (VAA) methodology [94] contributes by not only aiding in detection but also deepening insights into NLP model weaknesses. Finally, the DISP framework [95] offers a versatile and comprehensive solution for both detecting and mitigating malicious text perturbations, ensuring the integrity and security of textual data within NLP systems. Together, these diverse approaches constitute a formidable arsenal in the ongoing effort to secure NLP systems against word-level adversarial text attacks.

B. Related Work on Adversarial Training and Mitigation (Done)

In the realm of adversarial training and mitigation for text models, several notable approaches have emerged to enhance their resilience. The Textual Manifold-based Defense (TMD) [96], for instance, stands out with its two-step process combining language model fine-tuning with InfoGAN training, promising robust defence against adversarial attacks. SHIELD [97] offers versatility by focusing on bolstering model accuracy against black-box adversarial attacks. Emphasizing symbolic perturbations, Discrete Adversarial Training (DAT) [98] introduces a novel method to enhance model robustness. For specific attack vectors, Dirichlet Neighborhood Ensemble (DNE) [99] effectively addresses attacks relying on synonym substitutions. Further specialized techniques include Manifold Adversarial Augmentation (MAA) [100] for enhancing Neural Machine Translation (NMT) model robustness, and the Fast Gradient Projection Method (FGPM) [27] which offers an efficient defence against synonym substitution attacks. Broadening the scope, Character-Level Adversarial Training [101] adds another layer of security, while Robust Adversarial Training for Machine Reading Comprehension (MRC) [102] is tailored for MRC models. The FreeLB method [103] also leverages adversarial training to enhance the robustness and performance of language understanding models. Beyond these direct training modifications, other paradigms contribute significantly: dynamic red-teaming methodologies [104] highlight the role of human-model adversarial interactions in generating realistic attack patterns for more effective training. Prompt Adversarial Tuning [105] introduces a learnable guard-prefix to user prompts, jointly optimized against both adversarial and benign scenarios to achieve robust refusal behavior. Furthermore, offering provable guarantees, Erase-and-check [106] presents a certifiable defense based on token erasure, providing safety assurances under bounded adversarial modifications. These diverse strategies collectively underscore the promising directions in fortifying text models against adversarial threats.

C. Related Work on General Enhancement and Miscellaneous (Done)

In the realm of general enhancement and miscellaneous strategies for text model defence, a range of innovative approaches has emerged. Adversarial and Mixup Data Augmentation [107], for instance, stands as a fundamental technique for bolstering model robustness through data augmentation. BERT-Defense [108] demonstrates proficiency in countering orthographic adversarial attacks using BERT's probabilistic model, a valuable asset in mitigating such threats. The ADFAR Framework [109] further contributes to the defense arsenal by effectively combatting adversarial word substitution attacks. Concurrently, Frequency-Guided Word Substitutions (FGWS) [110] offer an efficient means to identify adversarial examples in NLP tasks based on word frequency properties. Robust Encodings (RobEn) [111] address adversarial typos without altering model architecture, while Iterative Training for Dialogue Safety [112] tackles the vital challenge of enhancing dialogue system resilience. Additionally, base-inflexion encoding [113] focuses on improving NLP accuracy for non-standard English dialects, thereby promoting inclusivity and accessibility in defended systems. Finally, in this group of established techniques, Robust Adversarial Augmentation (RAA) [114] holds promise in enhancing model robustness and generalization through high-level hidden representations.

More recent advancements introduce sophisticated frameworks and LLM-specific defenses. AegisLLM [115], for example, introduces a cooperative multi-agent framework for real-time defence against adversarial attacks without retraining, leveraging agentic optimization and inference-time adaptability. Understanding LLM capabilities in security also informs defense development, as highlighted by an empirical study from Shao et al. [116] evaluating LLMs' ability to solve cybersecurity Capture-The-Flag (CTF) challenges and benchmarking their automated problem-solving. Directly addressing online threats, AEGIS [117] presents an online adaptive content moderation system using an ensemble of LLM experts to enhance safety robustness dynamically. Comprehensive toolkits are also emerging; AISafetyLab [118] offers an integrated framework with attack, defence, and evaluation modules for

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AI safety, supporting diverse models and adversarial scenarios. Similarly, evaluation platforms like AiXamine [119] provide black-box testing across multiple aspects of LLM safety and security, uncovering vulnerabilities such as adversarial robustness issues. For alignment, SaRO [120] introduces a reasoning-based framework enhancing LLM safety through long-chain reasoning and safety-policy-driven reflection. Addressing specific vulnerabilities, BackdoorAlign [121] proposes a novel defence against fine-tuning-based jailbreak attacks by integrating secret-prompt-triggered safety examples. These diverse strategies collectively contribute to advancing text model defence, offering versatile tools against a broad spectrum of adversarial threats in NLP applications [122]. In conclusion, the continuous development of diverse and innovative defence strategies is essential for enhancing the security and reliability of NLP models against ever-evolving adversarial threats [123].

IV. Overview of Certified Robustness in NLP

V. Related Work on Dataset (Done)

Steffen Eger et al. [124] highlight the lack of realism in prevailing research scenarios compared to real-world applications like social networks, instead focusing on low-level role-based attacks that leverage human cognitive robustness and presenting a benchmark of nine distinct attack patterns to evaluate future NLP model resilience. Further highlighting evaluation challenges, Ailem et al. [125] have shown that performance correlations among prompts in major benchmarks are non-random, significantly impacting model rankings and calling for robustness checks that account for different distributional assumptions in LLM evaluations. To address such benchmark limitations, Dynabench was introduced by Kiela et al. [126] to intertwine dataset creation, model development, and evaluation, thereby tackling the tendency of models to excel on static benchmarks while faltering in real-world tasks. This vulnerability of evaluation systems is further underscored by findings from Raina et al. [127] that short universal adversarial phrases can mislead judge-LLMs in zero-shot evaluation settings, causing them to inflate scores—a critical issue demanding redesigned, robust evaluation systems.

The development of dynamic benchmarks, such as the NLI dataset by Nie et al. [128] collected through adversarial human-model interactions, aims to align with endless learning principles and avoid static test saturation. Extending this focus on dynamic and real-world challenges, the PowerPoint Task Completion-Robustness (PPTC-R) benchmark [129] evaluates large language models' ability to handle adversarial instructions across linguistic and software version changes, revealing systemic weaknesses in current models. Addressing dataset quality directly, Gardner et al. [130] introduced contrast sets to counter systematic dataset gaps that lead to misleading evaluations, proposing a rigorous annotation paradigm to ensure models capture intended capabilities rather than simplistic decision rules. Together, these works emphasize the critical need for realistic, dynamic, and comprehensive evaluation frameworks to truly advance the robustness and reliability of NLP models. Complementing these efforts, JailbreakBench serves as an evolving open-source framework for jailbreaking attack and defense research, featuring standardized evaluation metrics and a behavior dataset to enhance security-focused benchmarking [131]–[135].

In parallel, RUPBench [136] provides a comprehensive robustness evaluation across 15 reasoning datasets and nine types of textual perturbations, uncovering both model-size-dependent robustness patterns and persistent error types that call for architectural improvements. The critical role of data in these advanced evaluations is underscored by Gao et al. [137], who stress that high-quality annotated data remains vital for fine-tuning NLG evaluation models, particularly as LLMs expand task scenarios into instruction-response paradigms. However, as highlighted by Wu et al. [138], current limitations in dataset volume, complexity, and resource-intensive curation processes present major bottlenecks for developing effective detectors and evaluation metrics—challenges that demand innovative approaches to data resource design.

VI. Discussion and Conclusion

The rapid advancement of large language models (LLMs) has revolutionized natural language processing but has simultaneously exposed their vulnerability to sophisticated adversarial attacks, which pose significant risks to their reliability and security. This survey has provided a comprehensive review of textual adversarial attacks and defense mechanisms, categorizing attacks into sentence-, word-, and character-level manipulations, and detailing key defense strategies including detection, adversarial training, semantic analysis, and general system enhancements. The concept of certified robustness, which offers formal guarantees of model resilience, was also highlighted as a particularly promising avenue. Despite notable progress in understanding and mitigating these threats, substantial challenges persist concerning real-world applicability, scalability, generalization across diverse models and attack types, and the establishment of standardized benchmarking. As adversarial threats continue to evolve in complexity and subtlety, ongoing, focused research is crucial for the development of truly secure, reliable, and robust language models. This survey aims to serve as a valuable resource in this endeavor, fostering the creation of resilient NLP systems capable of withstanding increasingly sophisticated adversarial tactics.

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